# RETAIL DATA RESCUE

Excel & Power Query | Independent portfolio by Hakan

Data quality case study | Reference reporting with Excel audit queries

## Business problem

A retail transaction export mixes merchandise sales, cancellations, fees, adjustments and ambiguous records. The process makes rules and source-to-report reconciliations explicit.

## Prepared outputs

Excel reference dashboard with filters; complete M sources; immutable raw workbook; complete review register and reproducible classified export; decision log; quality audit; reconciliation and duplicate sensitivity; reproducible Python controls.

## Verified reference results

541,909 rows explained. Merchandise sales GBP 10,271,433.063; merchandise cancellations GBP 478,724.18; signed merchandise balance GBP 9,792,708.883. Missing customer IDs do not exclude eligible sales.

## Key decisions

Exact duplicates remain in the main result. The first-record-only alternative removes 5,268 rows and GBP 21,740.98 from raw signed transactions. Special codes are separated by explicit rules. C means cancellation, not confirmed physical return.

## Validation and acceptance

Python source checks, reconciliation and repeat/append tests passed. Microsoft M parser syntax checks passed. Four audit queries are embedded in the main workbook; source and missing-value refreshes were checked in Excel 2021. Full-model M sources are supplied separately; their end-to-end Excel refresh acceptance is pending.

## Limits

Internal source reconciliation only; no external accounting validation, cost/profit or tax conclusion. December 2010 observed dates end on 23 Dec; December 2011 ends on 9 Dec. Rankings and repeat scenarios need their stated scopes.

Source: Chen, D. (2015), Online Retail, UCI. DOI 10.24432/C5BW33. CC BY 4.0.

See docs/KURULUM_YENILEME_TR.md and validation/EXCEL_ACCEPTANCE_CHECKLIST.csv.